# D4.2 Stability and change (HL)

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 30 |

**About 30 minutes,** or one section at a time. The number of marks shows how many separate points to make.

## Section A: Recall

6 MARKS

1 **DEFINE** ecological succession. [D4.2.11] [1]

2 **DISTINGUISH** between primary and secondary succession. [D4.2.11] [2]

3 **STATE** one example of a pioneer organism. [D4.2.12] [1]

4 **DEFINE** climax community. [D4.2.14] [1]

5 **STATE** one human activity that arrests succession. [D4.2.14] [1]

## Section B: Understanding

11 MARKS

6 **EXPLAIN** how pioneer species make it possible for other species to grow. [D4.2.12] [3]

7 **OUTLINE** four changes that happen in a community during primary succession. [D4.2.12] [4]

8 **DESCRIBE** one example of cyclical succession. [D4.2.13] [2]

**9** **EXPLAIN** how grazing by livestock prevents a forest from developing. [D4.2.14] [2]

## Section C: Working with evidence

9 MARKS

*All data in this section are invented for practice. The patterns are realistic.*

**10** [D4.2.12]

Plots on lava flows of known age on a volcanic island were surveyed.

| AGE OF LAVA / YEARS | 0 | 10 | 50 | 100 | 300 |
|---|---|---|---|---|---|
| NUMBER OF PLANT SPECIES | 0 | 6 | 22 | 35 | 40 |
| SOIL DEPTH / CM | 0 | 0.5 | 4 | 15 | 40 |
| HEIGHT OF TALLEST PLANT / M | 0 | 0.1 | 1.5 | 8 | 25 |

**a** **DESCRIBE** the change in the number of plant species with age of lava. [2]

**b** **CALCULATE** the mean rate of soil formation between 50 and 100 years. Give the units. [2]

**c** **EXPLAIN** the relationship between soil depth and the height of the tallest plant. [2]

**d** **PREDICT** the number of plant species on lava 1000 years old. [1]

**e** **SUGGEST** how goats introduced at year 50 would have changed the later results. [2]

## Section D: Longer answer

4 MARKS

**11** **OUTLINE** the effects of draining a wetland on its community. [D4.2.14] [4]